2) Explain the Roles of operating systems, Compiler and assembler in Converting a Source program into executable program and enabling its execution.

⇒ Role of operating System:

The operating system manages Computer hardware and software resources. It loads the executable program into memory, allocate Resource and helps the program execute.

Compiler:

A Compiler translates a high level language program, such as C or C++, into assembly language or machine Code depending on the Compiler and Compilation process. It also detects syntax errors.

Assembler:

An assembler Converts an assembly language program into machine Code, which the processor Can understand and executes after req. linking steps.

Flow:

Source program → Compiler → Assembler → Linker → Executable program

Operating system. ←